

Alexander D. Knight

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Education

California Polytechnic State University (Cal Poly)	San Luis Obispo, CA
Bachelor of Science in Mechanical Engineering	Expected Graduation: June 2027
Overall GPA: 3.2	Dean's List: 1 quarter
Class Level: Junior	Cal Poly Scholars Program
Relevant Coursework: Calculus Solid Works MATLAB Dynamics Fluid Mechanics	

Technical Skills

SolidWorks | Fusion 360 | Machining | Soldering | Welding | MATLAB | 3D Printing | Laser Cutting
Water Jet | CNC/CAM | FPV Drones / Beta Flight | Microsoft Excel/Word/PowerPoint

Professional Experience

Research and Design Engineer

Vannon – San Luis Obispo, CA

August 2023 - Present

Design camper van accessories in CAD and then machine prototypes on a 5'x10' Laguna CNC router out of wood or aluminum. Make custom one-off brackets and mounts that go onto the custom van builds that are done in our shop. Conducted research and design work for running rails and roof racks that have now been put into production.

Machinist

Norgrove Gardens Railway – Arroyo Grande, CA

July 2026 – August 2026

Made and refurbished parts to restore 2ft gauge locomotives and rolling stock from Britain and South Africa to run wine tour trains at Norgrove Gardens

Design and Fabrication

Vintage Trolley Inc. – Lake Oswego, OR

June - August 2025

Designed and fabricated parts for converting a Portland streetcar from overhead to battery power to run on the Willamette Shore Trolley Line. Installed inverters and other high voltage components under the trolley car with mounting hardware that I designed and manufactured in house.

Intern Machinist

AMC Machining – Paso Robles, CA

June - August 2024

Shadowed employees setting up lathes and milling CNC machines. Wrote g-code for milling machines using Autodesk Fusion 360. Learned general production machine operations.

Sales Associate

Cambria Bicycle Outfitters – San Luis Obispo, CA

April 2021 - December 2023

Assisted customers in selecting bicycles, gear and parts. Performed repairs on customers bikes and assembled new bikes. Checked and labeled inventory.

Projects

32in Steel Gravel Bike

2026

Designed and built a gravel bike out of chromoly steel tubes using tig welding and metal 3D printing.

Voron 3D Printers

2024

Built two fully functional core xy 3d printers from open-source plans out of 3d printed parts.

Vannon V2 Running Rails

2023-24

Designed and fabricated prototype running rails for Mercedes Sprinter Vans. These are now installed on vans built out at Vannon.

CNC Milling Machine

2023

Designed and built a vertical CNC milling machine. Drew the plans and built it using the Haas CNC machines that I had access to at my high school. Won best in show at the California Mid State Fair for projects made in a high school shop.

FPV Drones

2020 - Present

Built, fly and configure first person view quad copters for filming action sports.

Model Airplanes

2019-21

Built countless radio control foam board airplanes to learn how model airplanes work and how to fly them. Built a balsa wood glider with an eight-foot wingspan.

7.5 Gauge Locomotive

2018

In middle school I built a 1/8 scale electric locomotive that my family could ride behind. I learned how to weld and built the whole thing myself. The first design had limited success, however after a few design revisions I built a locomotive that still runs well to this day.

Involvement and Leadership**Cal Poly Cycling Team - Road Race Director**

September 2023 - Present

Organize yearly USA Cycling sanctioned Cal Poly Criterium bike race. Attend out-of-town events and races. Support teammates at events and races.

Cal Poly Unicycle Club – Club Mechanic

September 2023 - Present

Logistics for out-of-state trips, teach new members how to ride and fix club-owned unicycles.

Cal Poly Future Fuels

September 2025 - Present

Fit electric drive systems to unusual vehicles including a couch.

NICA Coaching - Student Mountain Bike Coach

November 2021 - Present

Coach local High School Mountain Bike Team (SLO Scoters). Attend practice and connect with younger riders.

Cal Poly Juggling Club - Treasurer

September 2023 - June 2023

Submitted purchase requests for club equipment orders. Attended practice regularly.

Ffestiniog Railway Youth Volunteer Week

August 2022

Performed track and right of way maintenance as well as restored old rolling stock by removing rust and painting at Boston Lodge on the Ffestiniog narrow gauge railway in Wales.

San Luis Obispo Railroad Museum – Docent

January 2017 - June 2021

Showed visitors around, operated model railroad, created exhibits including a simulator.

Recognition and Awards**Cal Poly Dean's List Honors**

Fall 2023

Cal Poly Scholars Award Scholarship

2023

The Community Foundation San Luis Obispo

2023

Towru Kobara Scholarship for mechanical engineering students

Miner's Ace Hardware Scholarship

2023

King David's Masonic Lodge Scholarship

2023

California Mid State Fair

July 2023

Best in Show for Industrial Technology (CNC Milling Machine)

Presidential Volunteer Service Award – Gold

October 2018, 2019, 2020

Mayor's Community Service Award

April 2018, 2019

Interests

Mountain Biking | Road Cycling | Model Aviation | Unicycling | Railways | Sailing